THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + y = 9$ and $xy = 20$, then $x^2 + y^2 =$ 1

(a) 121 **(b)** 81 **(c)** 41 **(d)** 61

2. If $7x + 4y = 61$ and $4x + 7y = 54$, then $x - y =$ 1

(a) $\frac{7}{3}$ **(b)** 7 **(c)** $\frac{7}{11}$ **(d)** $\frac{115}{3}$

3. The pair $kx + 4y = k - 4$ and $16x + ky = k$ has no solution when $k =$ 1

(a) -8 **(b)** 16 **(c)** 8 **(d)** 4

4. The pair $kx + 8y = k - 8$ and $32x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 8 **(b)** 16 **(c)** -16 **(d)** 32

5. If $9x + 7y = 69$ and $4x + 5y = 42$, then $x + y =$ 1

(a) 6 **(b)** 9 **(c)** 3 **(d)** 18

6. The area (in sq. units) of the triangle formed by the line $5x + 2y = 20$ and the coordinate axes is: 1

(a) 20 **(b)** 2 **(c)** 40 **(d)** 200

7. If $x + y = 10$ and $xy = 21$, then $x^2 + y^2 =$ 1

(a) 58 **(b)** 79 **(c)** 100 **(d)** 142

8. If $6x + 2y = 38$ and $2x + 6y = 30$, then $x - y =$ 1

(a) 1 **(b)** 2 **(c)** 17 **(d)** 8

9. **Assertion (A):** The lines $3x + 2y = 5$ and $2x - 3y = 7$ are parallel. 1

Reason (R): A pair of linear equations is consistent if its lines intersect or coincide.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $x = 3,\ y = 2$ is the solution of $x + y = 5$ and $x - y = 1$. 1

Reason (R): A consistent pair of linear equations has at least one solution.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve: $7x + 4y = 61$ and $4x + 7y = 54$. **2**

12. Find the points where the line $4x + 5y = 20$ meets the coordinate axes and the area of the triangle so formed. **2**

13. Find k so that the pair $kx + 4y = k - 4$, $16x + ky = k$ has no solution. **2**

14. Solve: $6x + 2y = 38$ and $2x + 6y = 30$. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve: $\frac{2}{x} + \frac{5}{y} = \frac{33}{20}$ and $\frac{3}{x} + \frac{2}{y} = \frac{11}{10}$. **3**

16. Find a and b for which the pair $7x + y = 5$ and $(a + b)x + (a - b)y = 10$ has infinitely many solutions. **3**

17. Find the area of the triangle formed by the lines $x + 2y = 8$, $2x + y = 7$ and the x-axis. **3**

OR

Solve: $\frac{4}{x} + \frac{3}{y} = \frac{11}{6}$ and $\frac{2}{x} + \frac{5}{y} = \frac{3}{2}$. **3**

18. Find a and b for which the pair $4x + 2y = 3$ and $(a + b)x + (a - b)y = 9$ has infinitely many solutions. **3**

OR

Find the area of the triangle formed by the lines $3x + y = 9$, $x + y = 5$ and the x-axis. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A town planner models pairs of roads by linear equations. Pair A: $2x + 3y = 12$ and $4x + 6y = 24$. Pair B: $x + y = 5$ and $2x + 2y = 7$. Pair C: $x - y = 1$ and $x + y = 7$. **5**

(i) What is the geometrical relation between the roads of Pair A?

(ii) Do the roads of Pair B ever meet?

(iii) Find the junction where the roads of Pair C meet.

(iv) For what value of k are the roads $kx + 3y = 12$ and $4x + 6y = 24$ coincident? For which values of k do they meet at exactly one point?

20. A boat goes 45 km upstream and 60 km downstream in 13 hours, and 30 km upstream and 90 km downstream in 12 hours. Find the speed of the boat in still water and the speed of the stream. **5**

OR

Draw the graphs of $2x + y = 10$ and $x - y = 2$. Find the area of the triangle formed by these lines and the y-axis. **5**

END OF QUESTION PAPER — Total: 40 Marks